Feasibility Study

Feasibility is the measure of how practical and achievable a software project is. A feasibility study is conducted before the development of the software to determine whether the proposed system can be successfully implemented using the available resources, technology, budget, and time. It helps identify potential risks and ensures that the system will satisfy user requirements and organizational objectives.

The AI-Based Environmental Compliance Monitoring and Rule-Based Risk Prediction System has been evaluated from different perspectives to determine whether it is suitable for development. The proposed system aims to improve environmental compliance monitoring by replacing manual processes with a digital platform that provides automated compliance checking, rule-based risk prediction, dashboards, and report generation. The study confirms that the project is technically possible, economically affordable, and operationally beneficial.

The objectives of conducting the feasibility study are:

- To determine whether the proposed system satisfies the environmental monitoring requirements of industries.
- To verify that the system can be developed using the available technologies and development tools.
- To evaluate whether the project can be completed within the allocated budget and development schedule.
- To determine whether users can easily adopt the proposed system.
- To analyze the long-term benefits of implementing the system.
- To identify any technical or operational challenges before development begins.

Information Assessment

Before designing the system, information was collected from environmental officers, factory supervisors, and existing environmental monitoring practices. The following questions were considered during the feasibility study.

1. How would industries manage if the proposed system is not implemented?

Industries would continue using manual inspections, spreadsheets, and paper records to monitor environmental compliance. This would increase the chances of human errors,

delayed reporting, inefficient record management, and slow identification of environmental violations.

2. What problems exist in the current system?

The existing system depends heavily on manual data collection and record maintenance. Environmental data is scattered across different files, compliance checking requires manual comparison with environmental standards, reports consume considerable time to prepare, and risk identification occurs only after violations have already happened.

3. How will the proposed system contribute to organizational objectives?

The proposed system centralizes environmental monitoring, automates compliance verification, predicts compliance risks using predefined rules, generates reports instantly, improves monitoring efficiency, supports regulatory compliance, and helps management make informed decisions.

4. Can the system integrate with other organizational systems?

Yes. The proposed application is designed using a modular architecture and can later be integrated with IoT devices, cloud platforms, GIS systems, government compliance portals, and Machine Learning models for intelligent environmental monitoring.

5. Does the system require new technology?

The mini project uses widely available technologies such as Angular, ASP.NET Core, SQL Server, and Rule-Based Artificial Intelligence, which are stable and well supported. Future versions can include IoT sensors and Machine Learning without changing the overall architecture.

6. What functions must be supported?

The system should support user authentication, industry management, environmental monitoring, compliance checking, rule-based risk prediction, dashboard visualization, alert generation, historical record management, and report generation.

Types of Feasibility

1. Technical Feasibility

Technical feasibility determines whether the required hardware, software, and technical expertise are available for developing the proposed system.

The AI-Based Environmental Compliance Monitoring and Rule-Based Risk Prediction System can be successfully developed using existing technologies. The development

environment includes Angular for the frontend, ASP.NET Core for the backend, SQL Server for database management, and Visual Studio Code as the development platform. The rule-based prediction engine is implemented using predefined IF–THEN business rules, which are simple, reliable, and easy to maintain.

The development team possesses sufficient knowledge of web technologies, database management, and software engineering concepts required for implementation. The technologies selected are open-source or freely available for academic use, making the project technically practical. The system architecture also allows future integration of IoT sensors and Machine Learning models without requiring major modifications.

Technical Feasibility

Technical feasibility determines whether the required hardware, software, and technology are available to successfully develop and implement the proposed system.

The AI-Based Environmental Compliance Monitoring and Rule-Based Risk Prediction System (EnviroSafe) is technically feasible because it uses well-established technologies such as Angular for the frontend, ASP.NET Core for the backend, SQL Server for database management, and Visual Studio Code as the development environment. The rule-based prediction engine is implemented using predefined IF–THEN business rules.

The system also integrates low-cost IoT hardware, including the ESP32 microcontroller, DHT11 temperature and humidity sensor, and MQ-135 air quality sensor. These devices collect real-time environmental data and transmit it to the web application through Wi-Fi. The sensor readings are automatically stored in the database and compared with predefined environmental compliance standards. The selected IoT components are inexpensive, widely available, easy to program, and suitable for academic as well as prototype industrial applications.

The development team possesses the necessary knowledge of web development, database management, IoT programming, and software engineering. The chosen technologies are stable, reliable, and scalable, allowing future enhancements such as additional sensors, Machine Learning, cloud integration, and predictive analytics.

Technical Feasibility Analysis

- Uses Angular, ASP.NET Core, and SQL Server.
- Integrates ESP32, DHT11, and MQ-135 IoT devices.
- Supports real-time environmental monitoring through Wi-Fi.
- Uses low-cost and readily available hardware.

- Technologies are stable and well documented.
- Easy to extend with additional IoT sensors and Machine Learning.

Operational Feasibility

Operational feasibility evaluates whether the proposed system will function effectively in real-world industrial environments and whether users can easily operate it.

The proposed system simplifies environmental compliance monitoring by combining manual inspection data with real-time IoT sensor readings. Environmental inspectors can perform inspections while the IoT devices continuously monitor air quality, temperature, and humidity. The collected sensor data is automatically compared with predefined environmental standards, reducing manual effort and improving monitoring accuracy.

Administrators can manage industries, compliance standards, and users, while industry managers can monitor dashboards, review reports, and receive recommendations based on detected environmental risks. The user-friendly interface requires minimal training, making the system easy to adopt in industries.

Operational Feasibility Analysis

- Easy-to-use web interface.
- Supports both manual inspection and IoT-based monitoring.
- Reduces paperwork and manual calculations.
- Provides real-time environmental monitoring.
- Improves accuracy and decision-making.
- Suitable for industries and environmental regulatory authorities.

Economic Feasibility

Economic feasibility evaluates whether the expected benefits justify the cost of developing and implementing the proposed system.

The proposed system is economically feasible because it uses open-source software technologies and low-cost IoT components. Angular, ASP.NET Core, Visual Studio Code, and many development tools are freely available for academic use. The IoT hardware consists of affordable components such as the ESP32, DHT11, and MQ-135, with a total hardware cost of less than ₹1000, making the project cost-effective.

Although there is a small investment for IoT hardware, the system significantly reduces manual monitoring costs, improves environmental compliance, minimizes reporting effort, and helps industries avoid penalties caused by environmental violations. Therefore, the long-term benefits outweigh the implementation cost.

Economic Feasibility Analysis

- Uses free software development tools.
- IoT hardware cost is below ₹1000.
- Requires minimal hardware investment.
- Reduces manual monitoring and reporting costs.
- Improves productivity and compliance management.
- Provides significant long-term benefits with low implementation cost.